The Magic of Trapdoor Hash Function

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Disclaimer(s)

1. This is not my own work, just a research area I enjoy.

2. Goal: give an overview of this primitive and how to use it.

Trapdoor Hash Functions: Setting

- Alice only learns $f(x)$
- Bob learns nothing about f

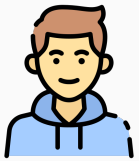
Trapdoor Hash Functions: Primitive

$\mathsf{Setup}(1^\lambda)$ $\mathsf{KeyGen}(\mathsf{hk}, f)$ $\mathsf{Hash}(\mathsf{hk}, x)$ $\mathsf{Encode}(\mathsf{ek}, x)$ $\mathsf{Decode}(\mathsf{td}, h)$

ek_f hk

$\mathsf{Decode}(td, h) = d$

h e



$h = \mathsf{Hash}(\mathsf{hk}, x)$

$e = \mathsf{Encode}(\mathsf{ek}_f, x)$

td f

x

Correctness*

$e = d \oplus f(x)$

Compactness

$|h| = \mathrm{poly}(\lambda)$

Rate-1

$|e| = |f(x)|$

Input Privacy*

Only $f(x)$ leaked

Function Hiding

$\mathsf{ek}_f \approx \mathsf{ek}_g$

TDH for Linear Functions

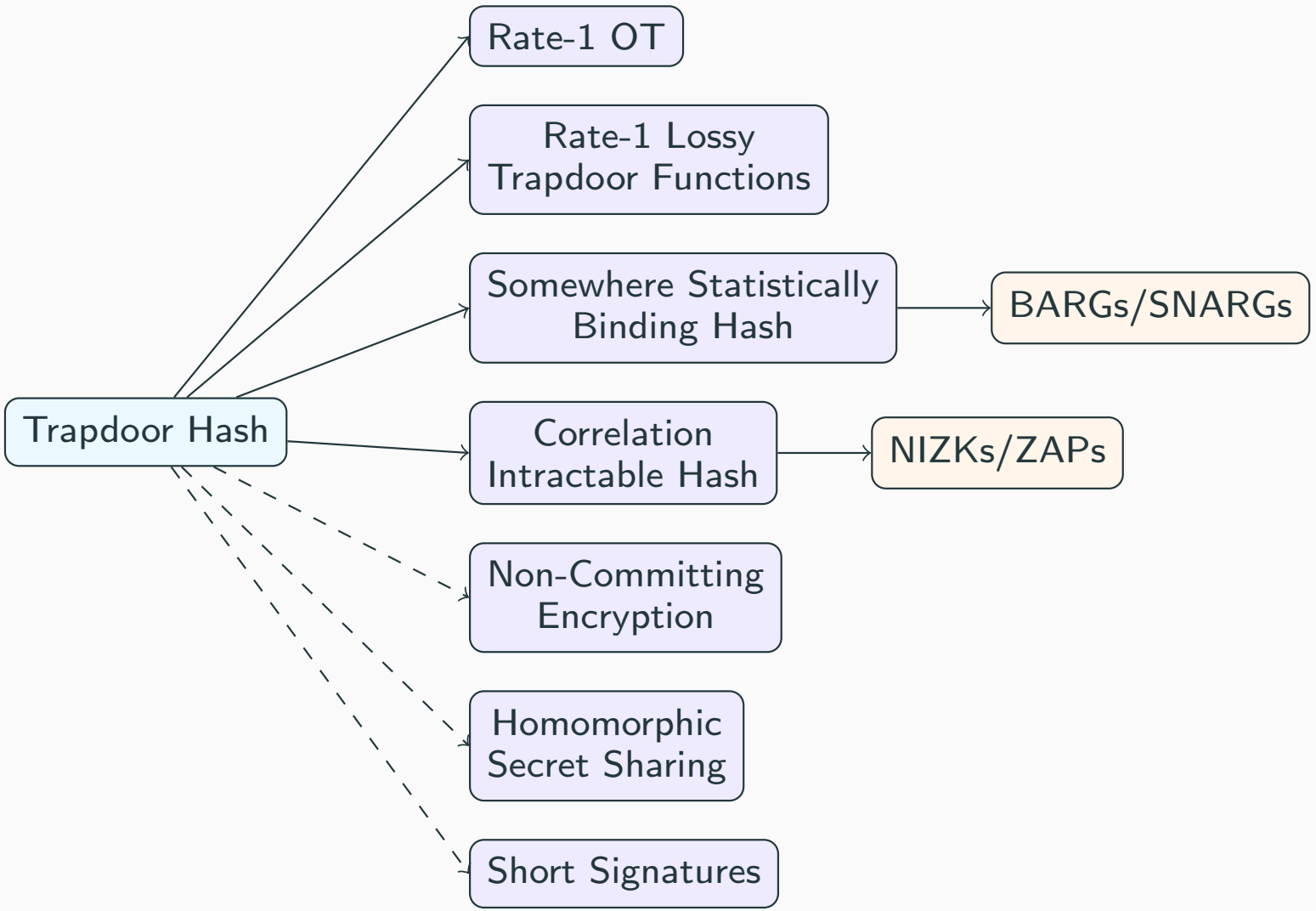
TDH supporting **linear functions** are known from:

- ✅ [DGI+19] Constructions under **LWE**, **DCR**, **QR**, **DDH**
- ❗ DDH,DCR and LWE with polynomial noise-to-modulus ratio suffer a $1/\text{poly}(\lambda)$ correctness error

- ✅ [AMR 25] Construction from low-noise **LPN**
- ❗ $1/\text{poly}(\lambda)$-error and low compactness

- ❌ No construction from Group Actions/Isogenies, MQ, Search Assumptions.

Applications



Applications In this talk

1. Semi-Honest Oblivious Transfer
2. Somewhere Statistically Binding Hash
3. Lossy Public Key Encryption
4. Correlation Intractable Hash

❗ All primitives above are **Rate-1**.

Oblivious Transfer

Oblivious Transfer

$b \in \{0, 1\}$

m_b

m_0, m_1

Sender Privacy

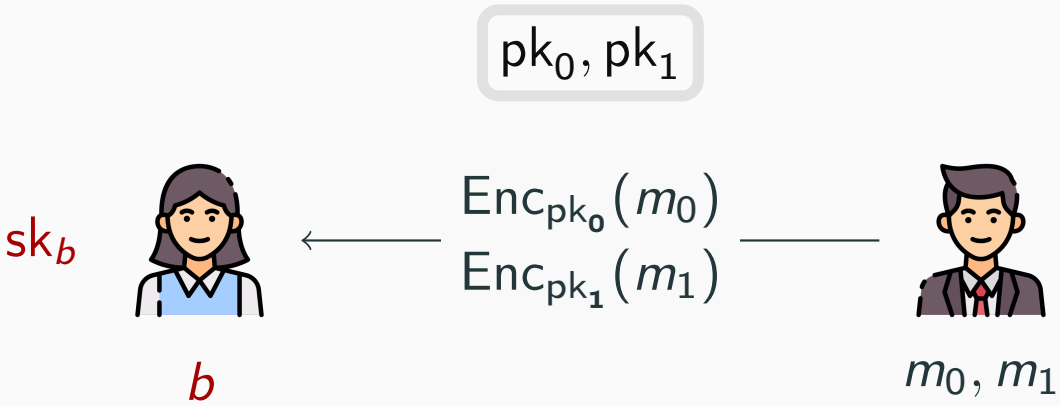
No leakage on m_{1-b}

Receiver Privacy

No leakage on b

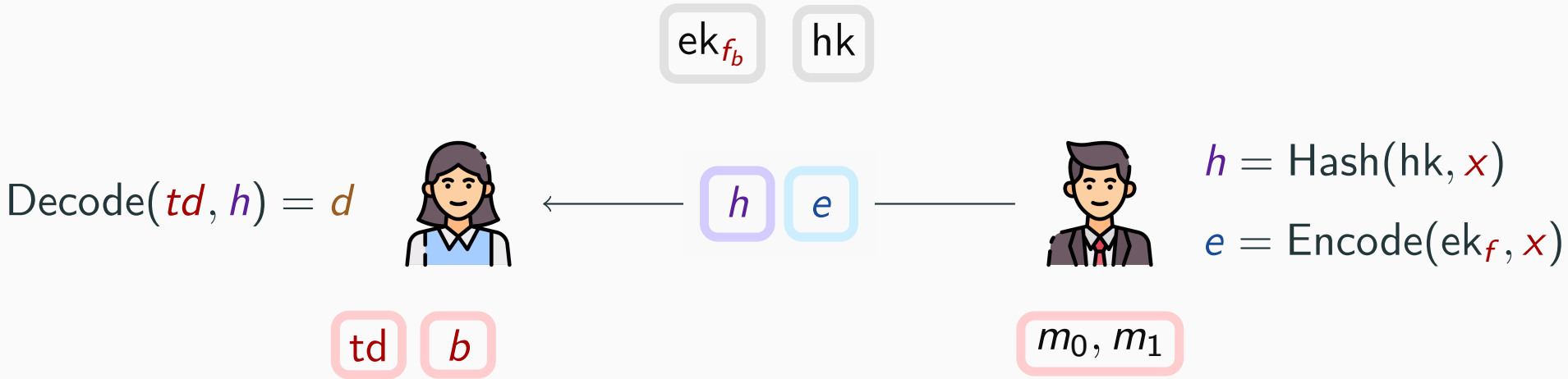
Rate-1/2 OT

Prior to TDH: **two-keys paradigm**



Inherently at best **Rate-1/2.**

OT from Linear Trapdoor Hash Functions



Function

$f_b(m_0, m_1) = m_b$

Correctness

$e = d \oplus m_b$

Rate ≈ 1

$|h| + |e| \approx |m_b|$

Batched OT from Linear Trapdoor Hash Functions

$\mathsf{ek}_{f_{\vec{b}}}$ hk

$\mathsf{Decode}(td, h) = d$ $\quad \mathsf{td}$ $\vec{b}$ $\quad \longleftarrow h \; e \longrightarrow$ $\quad \vec{m}_0, \vec{m}_1$

$h = \mathsf{Hash}(\mathsf{hk}, x)$

$e = \mathsf{Encode}(\mathsf{ek}_f, x)$

Function

$f_{\vec{b}}(\vec{m}_0, \vec{m}_1) = (m_{i,b_i})_{i=1}^n$

Correctness

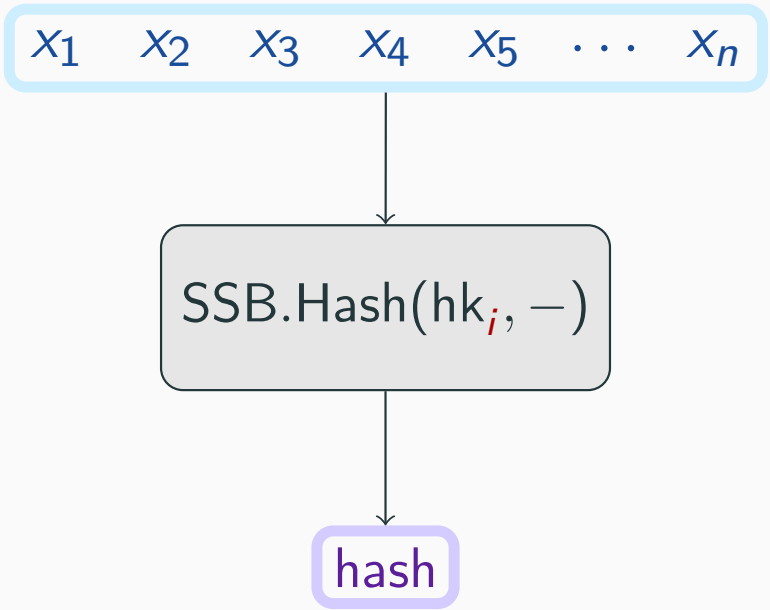
$e = d \oplus (m_{i,b_i})_{i=1}^n$

Rate ≈ 1

$|h| + |e| \approx \sum_i |m_{i,b_i}|$

Somewhere Statistically Binding Hash

SSB Hash



Somewhere Statistically Binding

Two inputs differing at the i-th position have different hash value

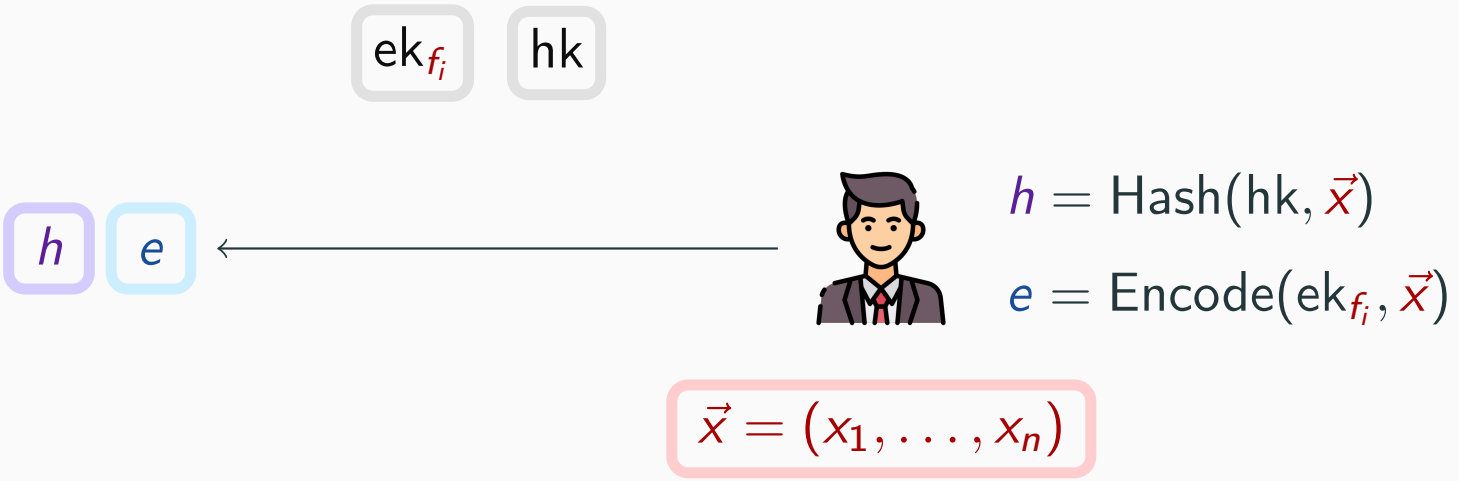
Index Hiding

$\mathsf{hk}_i \approx \mathsf{hk}_j$

Rate 1

$|\mathsf{hash}| \approx |x_i|$

SSB Hash from Linear Trapdoor Hash



Function

f_i the i-th projection

$f_i(\vec{x}) = x_i$

SSB

$e = d \oplus x_i$

(a reduction can extract x_i)

Rate ≈ 1

$|h| + |e| \approx |x_i|$

Lossy Public Key Encryption

Lossy Public Key Encryption

Normal Mode

$(\mathsf{pk}_1, \mathsf{sk}_1) \leftarrow \mathsf{GenInj}$

$m \mapsto \mathsf{Enc}(\mathsf{pk}_1, m; r)$
is injective/decryptable

$\overset{c}{\approx}$

Lossy Mode

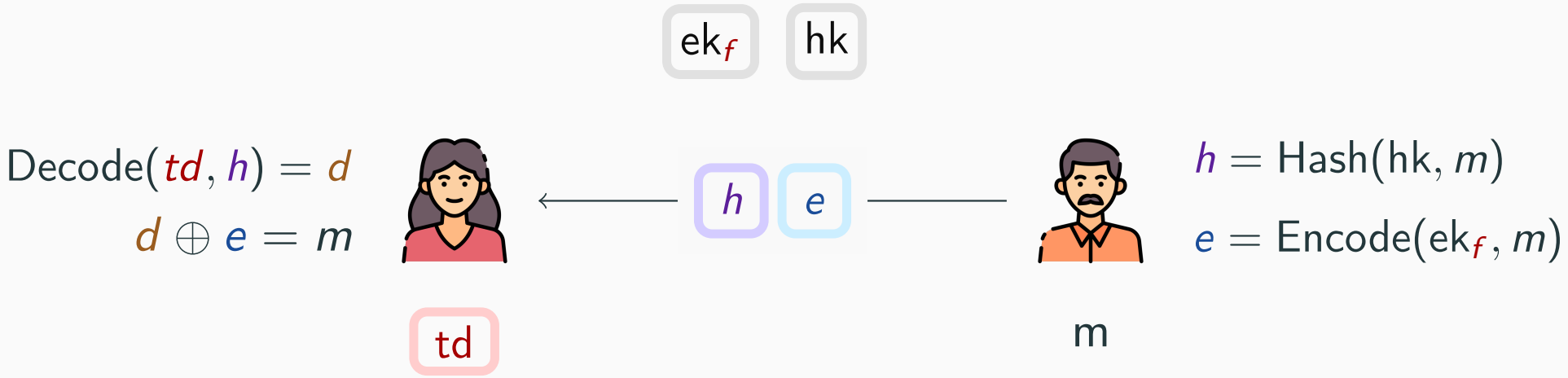
$\mathsf{pk}_0 \leftarrow \mathsf{GenLoss}$

$\log |\mathrm{Im}\, \mathsf{Enc}(\mathsf{pk}_0, -; r)| \ll |m|$
$\mathsf{Enc}(\mathsf{pk}_0, m)$ independent of m

✅ **Leakage Resilience**

✅ **Tight Security**

Encryption from Linear Trapdoor Hash



Function

$f(m) = m$

the identity

Correctness

$e = d \oplus m$

Rate-1

$|h| + |e| \approx |m|$

ciphertext is (h, e)

Encryption from Linear Trapdoor Hash

E.GenInj(1^λ):

$\mathsf{hk} = \mathsf{H.Setup}(1^\lambda, 1^n), \quad (\mathsf{td}, \mathsf{ek}_I) = \mathsf{H.KeyGen}(\mathsf{hk}, I)$

$\mathsf{pk} = (\mathsf{hk}, \mathsf{ek}_I), \quad \mathsf{sk} = \mathsf{td}$

E.GenLoss(1^λ):

$\mathsf{hk} = \mathsf{H.Setup}(1^\lambda, 1^n), \quad (\mathsf{td}, \mathsf{ek}_O) = \mathsf{H.KeyGen}(\mathsf{hk}, O)$

$\mathsf{pk} = (\mathsf{hk}, \mathsf{ek}_I)$

E.Encrypt(pk, m):

$h = \mathsf{H.Hash}(\mathsf{hk}, m), \quad e = \mathsf{H.Encode}(\mathsf{ek}_I, m)$

$\mathsf{ct} = (h, e)$

E.Decrypt$(\mathsf{pk}, \mathsf{ct})$:

$d = \mathsf{H.Decode}(\mathsf{td}, h)$

$m = e \oplus d$

Normal: $e = d \oplus m$

Lossy: $e = d$

Correlation Intractability

Correlation Intractable Hash

(Gen, Hash) is **correlation intractable** for $\mathcal{R}$ if given key $K \leftarrow \mathsf{Gen}(1^\lambda)$ it is hard to find $\vec{x}$ s.t.

$$(\vec{x}, \vec{y}) \in \mathcal{R} \quad \wedge \quad y_i = \mathsf{Hash}(K, x_i).$$

Examples

- **Collision Resistance**. $\mathcal{R}_{\mathsf{eq}}$ of (x_1, x_2, y_1, y_2) such that $x_1 \neq x_2$ and $y_1 = y_2$.
- **Near Collision Resistance**. $\mathcal{R}_{\mathsf{near}}$ of (x_1, x_2, y_1, y_2) such that $x_1 \neq x_2$ and $\Delta(y_1, y_2) \leq \delta$.
- **Search Relations.** $\mathcal{R}_C$ of (x, y) such that $y = C(x)$.

Correlation Intractability: Warm Up

Observation 1

CI for a **single** search relation $\mathcal{R}_C$ is easy:

$$x \mapsto C(x) \oplus r \qquad \text{for some } r \neq 0$$

... However, we aim to support search relations $\mathcal{R}_C$ for **all** $C \in \mathcal{C}$ some circuit family $\mathcal{C}$

Correlation Intractability: Warm Up

Observation 2

For a trapdoor hash, the encoding function for ek_C is shifted by a **very lossy function** from C.

$$e(x) = d(x) \oplus C(x)$$

$$d(x) = \mathsf{Decode}(td, \mathsf{Hash}(\mathsf{hk}, x))$$

$$\log |\mathrm{Im}\ d(x)| \leq \log |\mathrm{Im}\ \mathsf{Hash}(\mathsf{hk}, -)|$$

CI from Trapdoor Hash Function

We assume $C : \{0,1\}^n \to \{0,1\}^m$ has **large** output length $m \gg \lambda$.

CIH.Gen(1^λ):

$\mathsf{hk} = \mathsf{H.Setup}(1^\lambda, 1^n), \quad (\mathsf{td}, \mathsf{ek}_C) = \mathsf{H.KeyGen}(\mathsf{hk}, C), \quad \boldsymbol{r} \leftarrow^{\$} \{0,1\}^m$

Return $K = (\mathsf{ek}_C, \boldsymbol{r})$

CIH.Hash(K, x):

Return $\mathsf{hash} = \mathsf{H.Encode}(\mathsf{ek}_C, x) \oplus \boldsymbol{r}$

Corr. Intractability for C

$$e(x) \oplus \boldsymbol{r} = C(x)$$
$$\Rightarrow\ d(x) \oplus C(x) \oplus \boldsymbol{r} = C(x)$$
$$\Rightarrow\ \boldsymbol{r} = d(x) \quad \textit{(very unlikely)}$$

Corr. Intractability for $C' \neq C$

Use function hiding to switch

$\mathsf{ek}_C \approx \mathsf{ek}_{C'}$

Conclusion

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- TDHs are a versatile tool to build **succinct** primitives.
- Not known from some important assumptions (e.g. isogenies)
- Without LWE, not known to support expressive circuit families

Thanks for your attention!